

Abstract No.

Replication–Transcription Conflicts in Glioblastoma Stem-like Cells: Emerging therapeutic opportunities

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Glioblastoma (GBM) is the most prevalent and lethal primary malignancy of the adult brain. Although radiotherapy is the most effective non-surgical GBM therapy, recurrence is universal. A major driver of this recurrence is a highly tumor-initiating subpopulation of Glioblastoma Stem-like Cells (GSCs; CD133⁺), which survive treatment and repopulate the tumor. Therapeutic elimination of GSCs is difficult because they exploit core stress-response and self-renewal circuits that also support normal neural stem and progenitor cells. At the same time, the profound genomic instability of GBM and its structured cellular hierarchy—placing therapy-resistant GSCs at the top—make this disease an ideal system to identify and evaluate novel druggable vulnerabilities.

Here we show that patient-derived human GSC cultures exhibit pronounced replication stress and globally increased transcriptional activity. We propose that this dual burden elevates the incidence of Replication–Transcription Conflicts (RTCs) and DNA:RNA hybrids (R-Loops), triggering a heightened DNA Damage Response and fostering selection of aggressive, resistant GSC clones. Therefore, factors that mitigate RTC-associated lesions may provide targets to sensitize GSCs to therapeutic intervention.

In this study, we identify the homologous recombination protein RAD52 as a central component of the RTC damage-resolution network in GSCs. RAD52 takes part in pathways connected to conflict repair, including recombination-induced replication (BIR), which repairs and restarts collapsed replication forks, and transcription-associated homologous recombination repair (TA-HRR), which detects R-Loops and promotes faithful repair within highly transcribed loci. Based on these observations, our hypothesis points to RAD52 as a critical factor for GSC survival by promoting replication fork restart and clearing persistent R-Loops generated during RTCs, thereby limiting apoptosis and enabling resistance to radiotherapy. This motivates ongoing efforts to evaluate this protein inhibition, alone or combined with irradiation, as a strategy to suppress relapse while ideally sparing normal progenitors.



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